Current Status and Next Steps

Completed and Verified

- Data pipeline (EO loading, normalisation, ROI-based pairing) -built and verified.
- 4-class label scheme, corrected and verified (including the Wetlands correction).
- Balanced, disjoint 7-ROI train/val/test split -built and verified.
- Frozen-backbone baseline trained and properly evaluated on a held-out test set (52.51% → 53.57% test mIoU across iterations).
- Overfitting diagnosed and addressed via augmentation and early stopping.
- Land class failure diagnosed via visual inspection as a feature-representation issue, not a data-orientation issue.

In Progress

- LoRA fine-tuning of the Prithvi backbone combined with class-weighted loss, targeting the Land class specifically. First attempt (see Section 8) failed due to a NaN loss during training; the numerical cause needs to be debugged and the experiment re-run before results can be evaluated.

Not Yet Started

- EO + SAR multi-modal fusion.
- Physics-informed training using NDVI / NDWI / NDBI spectral indices.
- Spectral-index-based pseudo-labelling.
- Vectorisation of raster predictions into geo-referenced vector layers.
- Acquisition and integration of actual CARTOSAT imagery from the ISRO Bhoonidhi portal (reserved for final testing, per guide instruction).

Correction Cycle 5 — ESA WorldCover Ground-Truth Replacement

Rather than continuing to debug the NaN loss from the Correction Cycle 4 LoRA + class-weighted-loss run in isolation, a deeper audit was carried out first. This audit found a second, more serious problem sitting underneath the NaN instability: the fine-tuning checkpoint had been trained under a legacy label mode that mapped the Vegetation class (class 0) to the ignore index, meaning the model received zero training signal for Vegetation regardless of the loss instability. Combined, these two issues made the Correction Cycle 4 results unusable. Rather than patch this checkpoint, the decision was made to rebuild the ground-truth pipeline entirely using ESA WorldCover 10m labels (a 50x resolution improvement over the MODIS-derived 500m labels used until this point) and retrain from a fresh backbone and fresh LoRA adapters.

Ground-Truth Quality Audit

Before trusting the new WorldCover-based labels for training, the ground truth was independently audited across four dimensions:

Check	Result	Verdict
Spatial alignment (CRS, bounds, transform)	20 randomly sampled patches across all 7 ROIs matched exactly (0.0 m difference) between Sentinel-2 and WorldCover rasters	Passed

LUT / class-code completeness	All 11 raw ESA WorldCover codes (10–100) accounted for in the 4-class mapping; none silently dropped	Passed
Temporal alignment	WorldCover ground truth is dated 2020 (v100); Sentinel-2 imagery is dated 2017 — a 3-year gap with no matching 2017 WorldCover product available	Known limitation
Tile-stitch integrity	89 of 4,557 patches (54 in ROI 102, 35 in ROI 104) contain internal no-data seams from imperfect multi-tile mosaicking	Confirmed negligible impact

Note on the temporal gap:

this is a known, unavoidable limitation of pairing the fixed 2017 SEN12MS imagery with WorldCover (only available for 2020/2021). It is reported here as a documented caveat, not a resolved issue. Investigating Google Dynamic World (a per-scene, same-date land-cover product) as a replacement label source is listed under Next Steps below.

Retraining and Checkpoint Selection

- Backbone: Prithvi-EO-2.0-300M (pretrained, frozen — 303.89M parameters)
- LoRA adapters re-applied to self-attention projection layers (qkv, proj), rank r=16, α=32, dropout 0.1 — 17.48M trainable parameters (5.44% of total)
- Loss: log-smoothed inverse-class-frequency weighted CrossEntropyLoss ([0.672, 1.246, 1.134, 1.038]), ignore_index=255 — replacing the unweighted loss that produced the earlier NaN instability
- Optimizer: AdamW (lr=1e-4), CosineAnnealingLR, mixed precision (FP16), trained for 10 epochs on the corrected 7-ROI split

Per-epoch validation tracking (ROI 84) showed validation mIoU peaked at Epoch 2 (55.16%) and declined in every subsequent epoch, down to 40.43% by Epoch 10 — even as Vegetation IoU in isolation kept climbing to a peak of 92.41% at Epoch 7. This is the same overfitting signature identified in Correction Cycle 3, occurring again here: the model over-committed capacity to the dominant Vegetation class at the expense of the others, most visibly in Water recall, which fell from 96.6% (Epoch 2) to 25.8% (Epoch 10). The checkpoint saved as "best" (by validation mIoU) was directly verified against its saved metadata and confirmed to correctly correspond to Epoch 2, not a later, degraded epoch.

Final Test Results (Held-Out ROI 104, 508 Patches)

Class	IoU (%)	F1 (%)	Precision (%)	Recall (%)
Vegetation	66.87	80.15	74.68	86.48
Water	85.81	92.36	91.57	93.17
Built-up	46.14	63.14	51.16	82.47
Land	50.39	67.01	91.42	52.89

Overall Accuracy: 75.07% | Mean IoU: 62.30% | Test Loss: 0.5903

Re-evaluating on the 473 “clean” ROI 104 patches (excluding the 35 seam-affected ones) changed mIoU by only –0.12%, confirming the tile-stitch seams identified in the audit have a negligible effect on reported performance.

Land Class Diagnosis — Updated Confusion Matrix

Correction Cycle 3 diagnosed the Land-class failure via visual inspection as a feature-representation problem (Land confused with visually similar classes) rather than a data-orientation issue. The full confusion matrix from this retrained model confirms and quantifies that diagnosis precisely:

Ground Truth	Pred: Vegetation	Pred: Water	Pred: Built-up	Pred: Land
Vegetation	12,819,581	124,208	1,571,682	308,841
Water	97,156	2,616,140	30,925	63,724
Built-up	461,747	19,758	3,123,120	182,390
Land	3,787,016	96,970	1,379,702	5,909,766

Of all true Land pixels, 33.9% are misclassified as Vegetation — the model’s single largest confusion — versus 12.4% as Built-up and under 1% as Water. This is consistent with genuine spectral similarity between sparse cropland/bare soil and low-density vegetation, rather than a training or pipeline defect. Land precision remains high (91.4%): the model rarely over-predicts Land, it under-predicts it. This is an improvement over the Correction Cycle 3 baseline (where dedicated 100%-Land test patches scored 0–8% accuracy) but the underlying confusion is not fully resolved.

Visual Outputs — WorldCover-Retrained Model, Test Predictions (ROI 104)

Four held-out ROI 104 test patches, shown as Sentinel-2 RGB input, 4-class ground truth, and model prediction. The over-extension of Built-up (red) beyond its true boundaries and the loss of fine Land (pink) parcels into Vegetation are visible directly in patches 3 and 4, matching the confusion-matrix findings above.

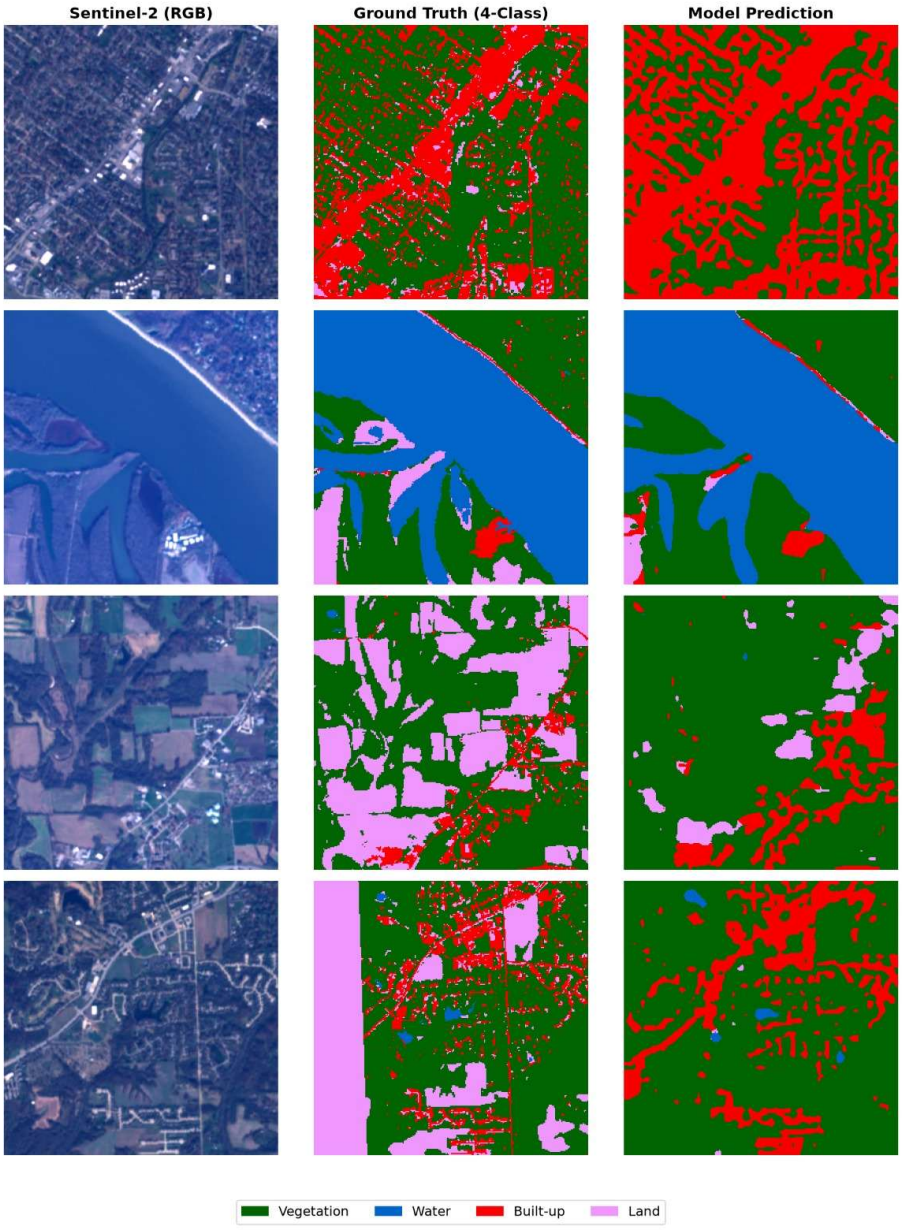


Figure 12. WorldCover-retrained model — test set prediction panels (Sentinel-2 RGB / Ground Truth / Prediction) across four ROI 104 patches, illustrating Built-up over-prediction and Land–Vegetation confusion.

Updated Status and Next Steps

Resolved Since Last Update

- The Correction Cycle 4 NaN-loss failure and its underlying corrupted-checkpoint cause (Vegetation silently mapped to ignore) — fixed via ground-truth pipeline rebuild and full retrain.
- Ground-truth pipeline independently audited for spatial alignment, class-code completeness, and tile-stitch integrity — all confirmed sound or confirmed negligible in impact.
- Checkpoint selection verified correct (Epoch 2, best validation mIoU) against saved metadata, ruling out use of an overfit later-epoch checkpoint.

- Code and notebooks pushed to GitHub for backup and reproducibility (data and checkpoints excluded via .gitignore).

Still Open

- Land–Vegetation confusion (33.9% of true Land pixels): improved from the Correction Cycle 3 baseline but not resolved — a genuine spectral-similarity limitation, not a pipeline defect.
- Built-up over-prediction (51.2% precision vs. 82.5% recall) and generally coarse, blocky prediction boundaries relative to ground truth, likely a consequence of the ViT's 16×16 patch tokenization.
- The 2017-imagery / 2020-label temporal gap remains unresolved within this dataset.

Not Yet Started

- Investigating Google Dynamic World as a same-date (per-scene) label source to eliminate the temporal gap without a full new imagery-acquisition effort.
- EO + SAR multi-modal fusion; physics-informed NDVI/NDWI/NDBI training; spectral-index pseudo-labelling; vectorisation into geo-referenced vector layers; CARTOSAT imagery integration (all as previously listed, still pending).